

Faculty Vitae of Jehanzeb Hameed Chaudhry

Name	Department	Date
Jehanzeb H. Chaudhry	Mathematics and Statistics	04/28/2020

Educational History

- Ph.D., August 2011, University of Illinois at Urbana-Champaign, Urbana, IL. Field of study: Computer Science (Certificate in Computational Science & Engineering. Dissertation title: Finite Element Methods for Implicit Solvent Models. Dissertation Advisor: Prof. Luke Olson.
- B.S., May 2003, Ghulam Ishaq Khan Institute of Engineering Sciences and Technology, Topi, Pakistan. Field of study: Computer Systems Engineering.

Employment History

- Assistant Professor, August 2015 – Present. Department of Mathematics and Statistics, The University of New Mexico, Albuquerque, NM.
- Postdoctoral Researcher, September 2014 – August-2015. Department of Scientific Computing (Appointment in lab of Prof. Max Gunzburger), Florida State University, Tallahassee, FL.
- Postdoctoral Researcher, August 2011 – August 2014. Department of Mathematics (Appointment in lab of Prof. Donald Estep), Colorado State University, Fort Collins, CO.

Description of Research Interests:

My main research area is numerical analysis and predictive computational science. My research focuses on developing and analyzing efficient and robust numerical methods and uncertainty quantification for the simulation of complex real-world systems. My research experience spans computational mathematics and computer science, and their applications to real-world multi-scale and multi-physics problems such as biomolecular electrostatics, electrical activity of the heart, chemical reactions, magnetohydrodynamics and oceanic flows. The importance of my research program is underscored by funding from the National Science Foundation and Sandia National Labs.

The main development tools used in my research are adjoint based a posteriori error estimation, the least-squares finite element method and reduced order modeling. These are powerful tools for the development and analysis of numerical methods for a variety of scientific and engineering problems. My recent research has focused on developing novel numerical methods and analyses which have significantly advanced the frontiers in these fields and are expected to have a major impact. Some of these recent advances include:

- (i) A posteriori error analysis for the time-parallel Parareal algorithm. This research enables efficient yet reliable solution of real-world problems on future's highly parallel machines and which takes into account all sources of numerical errors and quantifies the uncertainty in computed quantities-of-interest.
- (ii) Error estimation for the Poisson-Boltzmann equation which takes into account the regularity of the adjoint solution. The Poisson-Boltzmann equation is an important equation for understanding the electrostatic properties of biological molecules.

- (iii) A robust reduced order method which accounts for the error between the low order solution and the *true* solution. This is a major step forward in this field as previous such methods focused on error between the low order solution and a higher order (but not true) solution.
- (iv) An error analysis for first time to a threshold event (e.g. when a tsunami wave reaches a certain height). This is the first such analysis for quantities of interest which are not linear functionals (or ones which can be easily linearized) of the solution.

My current research and future research directions are focused on tackling emerging problems in computational science and uncertainty quantification. These include error analysis of the Shallow Water Equations, design of a reliable multilevel Monte Carlo approach for uncertainty quantification and domain decomposition methods for parallel computing.

Articles Published in Refereed Journals

- D. Bohra, J. H. Chaudhry, T. Burdyny, E. A. Pidko, and W. A. Smith. *Modeling the electrical double layer to understand the reaction environment in a CO₂ electrocatalytic system*. Energy & Environmental Science (2019)
- J. H. Chaudhry, J. N. Shadid, and T. Wildey. *A posteriori analysis of an IMEX entropy-viscosity formulation for hyperbolic conservation laws with dissipation*. Applied Numerical Mathematics, 135:129 (2019)
- J. H. Chaudhry. *A posteriori analysis and efficient refinement strategies for the Poisson–Boltzmann equation*. SIAM Journal on Scientific Computing, 40:A2519 (2018)
- J. H. Chaudhry, N. Burch, and D. Estep. *Efficient distribution estimation and uncertainty quantification for elliptic problems on domains with stochastic boundaries*. SIAM/ASA Journal on Uncertainty Quantification, 6:1127 (2018)
- J. Chaudhry, D. Estep, and M. Gunzburger. *Exploration of efficient reduced-order modeling and a posteriori error estimation*. International Journal for Numerical Methods in Engineering, 111:103 (2017)
- J. H. Chaudhry, J. Collins, and J. N. Shadid. *A posteriori error estimation for multi-stage Runge–Kutta IMEX schemes*. Applied Numerical Mathematics, 117:36 (2017)
- J. H. Chaudhry, D. Estep, S. Tavener, V. Carey, and J. Sandelin. *A posteriori error analysis of two-stage computation methods with application to efficient discretization and the Parareal algorithm*. SIAM Journal on Numerical Analysis, 54:2974 (2016)
- J. H. Chaudhry, D. Estep, V. Ginting, J. N. Shadid, and S. Tavener. *A posteriori error analysis of IMEX multi-step time integration methods for advection-diffusion-reaction equations*. Computer Methods in Applied Mechanics and Engineering, 285:730 (2015)
- J. H. Chaudhry, D. Estep, V. Ginting, and S. Tavener. *A posteriori analysis for iterative solvers for nonautonomous evolution problems*. SIAM/ASA J. Uncertainty Quantification, 3:434 (2015)
- A. Johansson, J. Chaudhry, V. Carey, D. Estep, V. Ginting, M. Larson, and S. Tavener. *Adaptive finite element solution of multiscale PDE-ODE systems*. Computer Methods in Applied Mechanics and Engineering, 287:150 (2015)

- A. N. Hirani, K. B. Nakshatrala, and J. H. Chaudhry. *Numerical method for Darcy flow derived using Discrete Exterior Calculus*. International Journal for Computational Methods in Engineering Science & Mechanics, 16:151 (2015)
- J. H. Chaudhry, E. C. Cyr, K. Liu, T. A. Manteuffel, L. N. Olson, and L. Tang. *Enhancing least-squares finite element methods through a quantity-of-interest*. SIAM J. Numer. Anal., 52:3085 (2014)
- J. H. Chaudhry, J. Comer, A. Aksimentiev, and L. N. Olson. *A stabilized finite element method for modified Poisson-Nernst-Planck equations to determine ion flow through a nanopore*. Communications in Computational Physics, 15:93–125 (2014)
- J. Chaudhry, D. Estep, V. Ginting, and S. Tavener. *A posteriori analysis of an iterative multi-discretization method for reaction-diffusion systems*. Computer Methods in Applied Mechanics and Engineering, 267:1 (2013)
- J. H. Chaudhry, S. D. Bond, and L. N. Olson. *A weighted adaptive least-squares finite element method for the Poisson-Boltzmann equation*. Applied Mathematics and Computation, 218:4892 (2012)
- J. H. Chaudhry, S. D. Bond, and L. N. Olson. *Finite element approximation to a finite-size modified Poisson-Boltzmann equation*. Journal of Scientific Computing, 47:347 (2011)
- S. D. Bond, J. H. Chaudhry, E. C. Cyr, and L. N. Olson. *A first-order system least-squares finite element method for the Poisson-Boltzmann equation*. Journal of Computational Chemistry, 31:1625 (2010)

Submitted for publication

- J. H. Chaudhry, L. Olson, and P. Sentz. *A least-squares finite element reduced basis method*. SIAM Journal on Scientific Computing
- J. H. Chaudhry, D. Estep and S. Tavener. *A posteriori error analysis for schwarz overlapping domain decomposition methods*. Computer Methods in Applied Mechanics and Engineering
- J. H. Chaudhry and J. Collins. *A posteriori analysis of the spectral deferred correction method*. Journal of Computational and Applied Mathematics
- J. Chaudhry, D. Estep, Z. Stevens, and S. Tavener. *Error estimation and uncertainty quantification for first time to a threshold value*. BIT Numerical Mathematics

In preparation

- J. H. Chaudhry, C. Cooper and V. Ramm. *Efficient mesh refinement for the Poisson-Boltzmann equation with boundary elements*.
- J. H. Chaudhry, A. Rappaport and J. Shadid. *An a posteriori error analysis for the equations of stationary incompressible magnetohydrodynamics*.

Presentations at Professional Meetings

- *A posteriori error analysis of overlapping domain decomposition techniques.* Conference on Computational Mathematics and Applications, University of Nevada, Las Vegas, October 2019.
- *Error Estimation for Multiplicative Schwarz Domain Decomposition Method.* SIAM Conference on Computational Science and Engineering (CSE19), Spokane, WA, February 2019.
- *A Posteriori Analysis and Efficient Refinement Strategies for the Poisson-Boltzmann Equation.* The 13th World Congress in Computational Mechanics, New York City, NY, July 2018.
- *A Posteriori Analysis of Implicit-Explicit (IMEX) Methods for Hyperbolic Systems.* The 3rd Annual Meeting of SIAM Central States Section, Fort Collins, CO, September 2017.
- *Efficient Distribution Estimation and Uncertainty Quantification for Elliptic Problems on Domains with Stochastic Boundaries.* 14th U.S. National Congress on Computational Mechanics (USNCCM14), Montreal, July 2017.
- *A Posteriori Error Analysis of the Parareal Algorithm.* SIAM Conference on Computational Science and Engineering (CSE17), Atlanta, Georgia, February 27 – March 3, 2017.
- *Density Estimation for a Class of Elliptic Problems on Stochastically Perturbed Domains.* SIAM Annual Meeting, Boston, MA, July 11 – July 16, 2016.
- *Density Estimation for a Class of Elliptic Problems on Stochastically Perturbed Domains.* ECCOMAS Congress 2016, Crete, Greece, June 5 – June 10, 2016.
- *A Posteriori Analysis of Implicit-Explicit (IMEX) Methods.* 13th U.S. National Congress on Computational Mechanics, San Diego, CA, July 27 – July 30, 2015.
- *A Posteriori Analysis of the Parareal Algorithm: Efficient Resource Allocation and Convergence Criteria.* SIAM Conference on Computational Science and Engineering (CSE15), Salt Lake City, Utah, March 14 – March 18, 2015.
- *Quantity-of-Interest Based Least-Squares Finite Element Methods.* SIAM Conference on Computational Science and Engineering (CSE15), Salt Lake City, Utah, March 14 – March 18, 2015.
- *Multi-level Parallel Time Integration.* 13th Copper Mountain Conference on Iterative Methods, Copper Mountain, Colorado, April 6 – April 11, 2014.
- *Goal-oriented least-squares finite element methods.* 12th U.S. National Congress on Computational Mechanics, Raleigh, North Carolina July 24, 2013.
- *A posteriori analysis of implicit-explicit methods.* SIAM Conference on Computational Science & Engineering (CSE13), Boston MA, February 27, 2013.
- *An a posteriori analysis of iterative technique for system of ODEs.* Joint Mathematics Meetings, San Diego, CA. January 12, 2013.
- *A posteriori analysis of multirate and multiscale evolution systems.* SIAM Annual Meeting, Minneapolis, MN. July 11, 2012.

- *Finite element method for the modified Poisson-Nernst-Planck equations (MPNPE)*. SIAM Conference on Computational Science and Engineering (CSE11), Reno, Nevada. February 28, 2011.

Colloquium and seminar talks at Universities Other than UNM

- *Efficient Distribution Estimation and Uncertainty Quantification for Elliptic Problems on Domains with Stochastic Boundaries*. Tufts University, September 12, 2019.
- *Efficient Distribution Estimation and Uncertainty Quantification for Elliptic Problems on Domains with Stochastic Boundaries*. University of Colorado-Denver, March 13, 2017.

Colloquium and seminar talks at UNM

- *A posteriori Error Estimation for Schwarz Domain Decomposition Methods*. Applied Math Seminar, April 1, 2019.
- *A Posteriori Error Analysis of the Poisson-Boltzmann Equation*. Department Colloquium, November 16, 2017.
- *A posteriori Error Estimation for Schwarz Domain Decomposition Methods*. Applied Math Seminar, April 1, 2019.
- *QoI based Finite Element Methods*. Applied Math Seminar, September 14, 2015.
- *Adjoint based A posteriori Error Estimation*. Applied Math Seminar, September 21, 2015.

Research Funding

- Title: “*Collaborative Research: A posteriori Error Analysis for Complex Models with Applications to Efficient Numerical Solution and Uncertainty Quantification*”. Principal investigator: Jehanzeb H. Chaudhry. Funding organization: National Science Foundation. NSF-DMS-1720402 (Computational Mathematics). August 2017 – July 2020. Amount awarded \$100,160.
- Title: “*A Parallel-in-Time Multigrid Methods for Hyperbolic Problems, with a Focus on the Shallow Water Equations*”. Co-PI sub-contract through Lawrence Livermore National Laboratory, Livermore, CA. June 2017. Total Award Amount \$69,971.
- Title: “*Adjoint error estimation for Entropy viscosity formulations for Euler Equations*”. Sub-contract through Sandia National Laboratories, Albuquerque, NM. June 2017. Amount \$17,500.
- Title: “*Adjoint based analysis and verification for Implicit-Explicit Schemes for Shock-Hydrodynamics*”. Principal investigator: Jehanzeb H. Chaudhry (with John Shadid and Tim Wildey at Sandia National Laboratories). Funding organization: Sandia National Laboratories: Laboratory Directed Research and Development (LDRD) Funding under Academic Alliance Program. February 2016 – September 2016. Amount awarded: \$50,000.

Doctoral Advisement

- Zachary Stevens. May 2021 (expected). Tentative title: *Error Estimation and Uncertainty Quantification for first time to a threshold value*.
- Trevor Giannini. August 2021 (expected). Tentative title: *A posteriori error estimation for Shallow Water Equations*.

Masters Advisement

- Ari Rappaport. August 2020 (expected). Title: *A posteriori error analysis for Magnetohydrodynamics (MHD)*.
- Nicholas Abel. August 2020 (expected). Title: *Multilevel Asymptotic Parallel-in-Time Techniques for Temporally Oscillatory PDEs*.

Bachelor's Honors Advisement

- Ari Rappaport. August 2017-May 2018. BS in Mathematics. Title: *Exploration of Black Box Multigrid for Linear Systems Modeling Resistive MHD*.

Undergraduate Student Mentoring

- Ari Rappaport. Spring 2017, Fall 2017 and Spring 2018. Independent study on finite element methods and multigrid algorithms.
- Christopher Leap. Spring 2019. Independent study on machine learning for predicting number of conjugate gradient iterations for a linear system.

Classroom Teaching

Year	Semester	Name of course	Course number	# students
2015	Fall	Introduction to Mathematical Thinking and Discrete Structures	Math 327	27
2016	Spring	Linear Algebra	Math 321	42
2016	Spring	Intro Numerical Computing	Math/CS 375	23
2016	Fall	Numerical Partial Differential Equations	Math 578	9
2016	Fall	Intro Numerical Computing	Math/CS 375	20
2017	Spring	Intro to Numerical Linear Algebra	Math 504	10
2017	Spring	Intro Numerical Computing	Math/CS 375	23
2017	Fall	Linear Algebra	Math 321	22
2017	Fall	Introduction to Numerical Approximation and Differential Equations	Math/CS 505	8
2018	Fall	Intro Numerical Computing	Math/CS 375	36
2018	Fall	Numerical Partial Differential Equations	Math 578	7
2019	Spring	Intro Numerical Computing	Math/CS 375	26
2019	Fall	Intro Numerical Computing	Math/CS 375	25
2020	Spring	Numerical Ordinary Differential Equations	Math 579	10

Service

Reviewing for journals

- Refereed an article for Mathematics of Computation (2020).
- Refereed two articles for International Journal for Numerical Methods in Engineering (2018 and 2020).
- Refereed two articles for SIAM Journal on Scientific Computing (2016 and 2018).
- Refereed an article for International Journal of Computer Mathematics (2018).
- Refereed an article for Applied Mathematics Modelling (2017).
- Refereed an article for IEEE Journal on Computational Electronics (2016).
- Refereed two articles for SIAM Journal on Numerical Mathematics (2015 and 2017).
- Refereed an article for Rocky Mountain Journal of Mathematics (2015).
- Refereed an article for Communications in Mathematical Sciences (2015).

Reviewing for national funding organizations

- Served on an NSF Panel for reviewing applications for the DMS division in March 2019. I was the primary reviewer for 9 proposals.
- Refereed a research proposal for Research Grants Council (RGC) of Hong Kong (2017).
- Refereed a research proposal for King Abdullah University of Science and Technology (2015).

Administrative work in Department, College, University committees

- Co-organizer of the statewide UNM-PNM Math Contest. 2018 — Present.
- Member of the Curriculum Committee. Fall 2019 — Present.
- Produced paperwork and other tasks to get Math 577 back into the catalog.
- Served on the team to make Numerical Analysis qualifying exam in Fall 2017, Spring 2018 and Fall 2019.
- Chair of the department Scheduling Committee. Fall 2017 — 2018.
- Member of the department Scheduling Committee. Fall 2016 — Fall 2017.
- Member of the department Hiring Committee. Fall 2017.
- Masters thesis committee member for Masumi Sugiyama. Fall 2019.
- PhD Comprehensive Exam committee member for Shu Wang, Fall 2019.
- Master's thesis committee member for Michael Bredemann. Spring 2020.

Community service, etc.

- Organized a two part minisymposium at SIAM Conference on Computational Science and Engineering (CSE19), Spokane, WA, February 25 – March 1, 2019.
- Organized a two part minisymposium at the 3rd Annual Meeting of SIAM Central States Section, Fort Collins, Colorado, September 29 – October 1, 2017.
- Organized a two part minisymposium at SIAM Conference on Computational Science and Engineering (CSE17), Atlanta, Georgia, February 27 – March 3, 2017.
- Helped the students establish the Society of Applied and Industrial Mathematics (SIAM) student chapter at UNM.

- Chaperoned around ten graduate students to a student conference organized at Texas Tech University, Lubbock, Texas.
- Helped grade the student answers for the 2015 UNM-PNM math contest.
- Reviewed part of the Academic Program Review document for the Mathematics and Statistics department.
- Jointly prepared and graded the Numerical Analysis qualifying exam in Fall 2017.